

So Meta

⚠ **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

Find the flag in this picture.

Hints

- What does meta mean in the context of files?
 - Ever heard of metadata?
-

Tools Used

- `file` – Determine file type
 - `exiftool` – Read and extract metadata from image files
-

Complete Solution

Step 1: Download and Prepare

Download the file named `pico_img.png` from the challenge page.



At first glance, it's a simple image – but the flag isn't visible in the picture itself.

Step 2: Basic File Inspection

First, verify the file type:

```
file pico_img.png
```

Expected output:

```
pico_img.png: PNG image data, 600 x 600, 8-bit/color RGB, non-interlaced
```

The file is a valid PNG image.

Step 3: Understand the Hint

The hint asks: *"What does meta mean in the context of files?"* and mentions *"metadata"*.

Metadata is data about data – information embedded within a file that describes its properties, origin, author, creation date, and more. For images, metadata can include camera settings, GPS coordinates, software used, and even hidden comments.

Step 4: Examine Metadata with exiftool

Use `exiftool` to extract all metadata from the image:

```
exiftool pico_img.png
```

Output:

```
ExifTool Version Number      : 13.50
File Name                    : pico_img.png
Directory                   : .
File Size                    : 109 kB
File Modification Date/Time   : 2026:05:20 17:56:49+02:00
File Access Date/Time        : 2026:05:20 17:56:55+02:00
File Inode Change Date/Time   : 2026:05:20 17:56:55+02:00
File Permissions              : -rw-rw-rw-
File Type                    : PNG
File Type Extension          : png
MIME Type                    : image/png
Image Width                  : 600
Image Height                 : 600
Bit Depth                    : 8
Color Type                   : RGB
Compression                  : Deflate/Inflate
Filter                       : Adaptive
Interlace                    : Noninterlaced
Software                     : Adobe ImageReady
XMP Toolkit                  : Adobe XMP Core 5.3-c011 66.145661, 2012/02/06-
14:56:27
Creator Tool                  : Adobe Photoshop CS6 (Windows)
Instance ID                  : xmp.iid:A5566E73B2B811E8BC7F9A4303DF1F9B
Document ID                  : xmp.did:A5566E74B2B811E8BC7F9A4303DF1F9B
Derived From Instance ID     : xmp.iid:A5566E71B2B811E8BC7F9A4303DF1F9B
Derived From Document ID     : xmp.did:A5566E72B2B811E8BC7F9A4303DF1F9B
```

Artist	: picoCTF{<REDACTED>}
Image Size	: 600x600
Megapixels	: 0.360

Key finding: The `Artist` field contains the flag!

```
Artist : picoCTF{<REDACTED>}
```

One-Liner Solution

You can extract the flag directly with a single command:

```
exiftool -Artist pico_img.png
```

Output:

```
Artist : picoCTF{<REDACTED>}
```

Or using `grep` to filter:

```
exiftool pico_img.png | grep Artist
```

Final Result

```
picoCTF{<REDACTED>}
```

Note: The actual flag is redacted here. In the real challenge, running `exiftool` will reveal the complete flag in the `Artist` field.

Key Concepts

Concept	Description
Metadata	Data about data – information embedded within a file that describes its properties.
EXIF (Exchangeable Image File Format)	A standard for storing metadata in image files (camera settings, GPS, copyright, etc.).
PNG Metadata	PNG files can store metadata in text chunks (tEXt, zTXt, iTXt).
exiftool	A powerful command-line tool for reading, writing, and editing metadata across many file formats.
Artist Field	A common metadata field that stores the creator or author of the image.

Pro Tips

- Metadata is often overlooked** – Many people forget to clean metadata before sharing files. This is a common source of flags in CTF challenges.
- exiftool is essential** – It’s the most comprehensive metadata tool available. Install it with `sudo apt install exiftool` or `brew install exiftool`.
- Check all metadata fields** – Flags can hide in `Artist`, `Comment`, `Copyright`, `Description`, `Title`, `XMP`, or custom fields.
- Use `-a` to see duplicate tags** – Some files have multiple instances of the same tag. `exiftool -a` shows all of them.
- Extract specific fields** – `exiftool -Artist file.jpg` returns only the Artist field, useful for scripting.
- Even empty fields matter** – Sometimes the absence of expected metadata can be a clue.

Common Metadata Fields to Check

Field	Description
Artist	Creator of the image
Comment	User comments
Copyright	Copyright information
Description	Image description
Title	Image title
XMP	Extended metadata platform
DocumentID	Unique document identifier
CreatorTool	Software used to create the image